



BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

Homework Assignment 1

【Spring Semester, 2025】

Due Date: March 12, 2026

Problem 1: (50%)

Rakata is a small island between the islands of Sumatra and Java in the South Pacific. It is famous for being the largest remnant of Krakatau Island after the notorious 1883 eruption. Whittaker et al. (1989) and Thornton et al. (1993) compiled historical plant and animal surveys of Rakata from 1886 to 1992; the approximate data for vascular plants species numbers (R), immigration rate (I) and extinction rate (E) are:

R	0	36	80	155	210	240
I	8.0	3.0	5.0	6.5	4.0	2.5
E	0.0	0.05	0.10	0.50	1.75	1.75

- (a) Use linear regression to estimate the immigration and extinction rates.
- (b) Re-write Eq. 1.2 using these Rakata data.
- (c) Estimate the equilibrium number of vascular plant species on Rakata.
- (d) How many species are in the mainland pool?
- (e) Write a computer program (C++, python, or MATLAB) based on recursive finite difference equation to simulate the species dynamics using the parameters you estimated and starting with no plants. Also simulate a scenario representing the pre-explosion condition in which the initial number of plant species is 500. Assuming only that the island size changed, how long would it take to achieve the current projected equilibrium level of about 250 species?

References:

- Thornton, I. W. B., R. A. Zann, and S. van Balen. 1993. Colonization of Rakata (Krakatau Is.) by non-migrant land birds from 1883 to 1992 and implications for the value of island equilibrium theory. *Journal of Biogeography* 20:441-452.
- Whittaker, R. J., M. B. Bush, and K. Richards. 1989. Plant recolonization and vegetation succession on the Krakatau Islands, Indonesia. *Ecological Monographs* 59:59-123.



Simberloff, D. S., and E. O. Wilson. 1970. Experimental zoogeography of islands. A two-year record of colonization. *Ecology* 51(5): 934-937.

Problem 2: (50%)

Design an alternative model for the island biogeography situation that uses curvilinear immigration and extinction functions. Consider a negative exponential and simple quadratic, respectively.

- (a) Graph the new rates of change against R and qualitatively sketch the dynamics of colonization from an empty island. Contrast these dynamics with those of the original model.
- (b) Write a new finite-difference equation and show that the equilibrium number of species satisfies

$$\frac{I_x}{E_x} = R^2 e^{aR}$$

- (c) Speculate on a biological mechanism that might support this alternative.
- (d) Modify the model to explicitly incorporate the effect of island size on immigration and extinction rates. Develop a size-dependent formulation that accounts for the increased habitat availability and reduced extinction risk associated with larger islands. Update the finite-difference equation accordingly and discuss the expected impact of island size on equilibrium species richness and colonization dynamics.

Notes:

1. Please submit your homework report together with your program to the NTU COOL course website.
2. Late submission will have a penalty of 10% discount per day of your homework total score toward a maximum of 50% discount. No late submission over five days will be accepted.